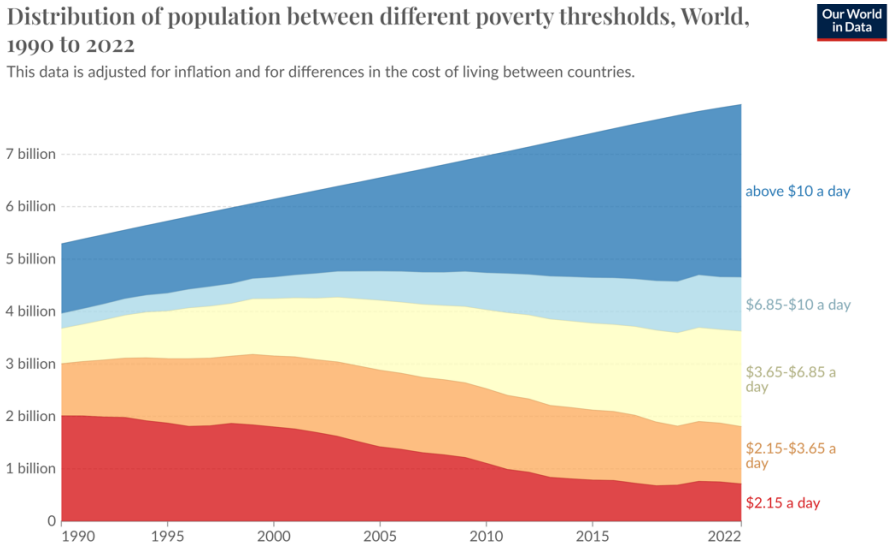
Shortfall of Domestic Resources to Eradicate Extreme Poverty by 2030

Adrien Fabre (CNRS, CIRED)

August 2024

Introduction

Poverty eradication

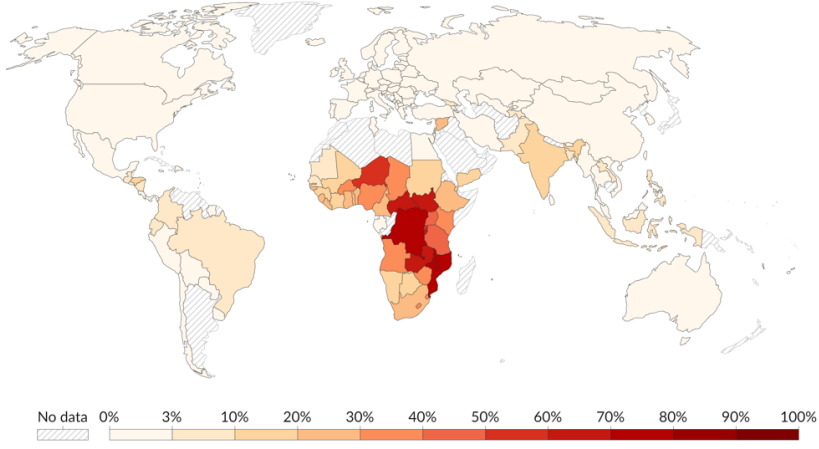


Poverty eradication

Share of population living in extreme poverty, 2019

Our World in Data

Extreme poverty is defined as living below the International Poverty Line of $2.15 per day. This data is adjusted for inflation and for differences in the cost of living between countries.

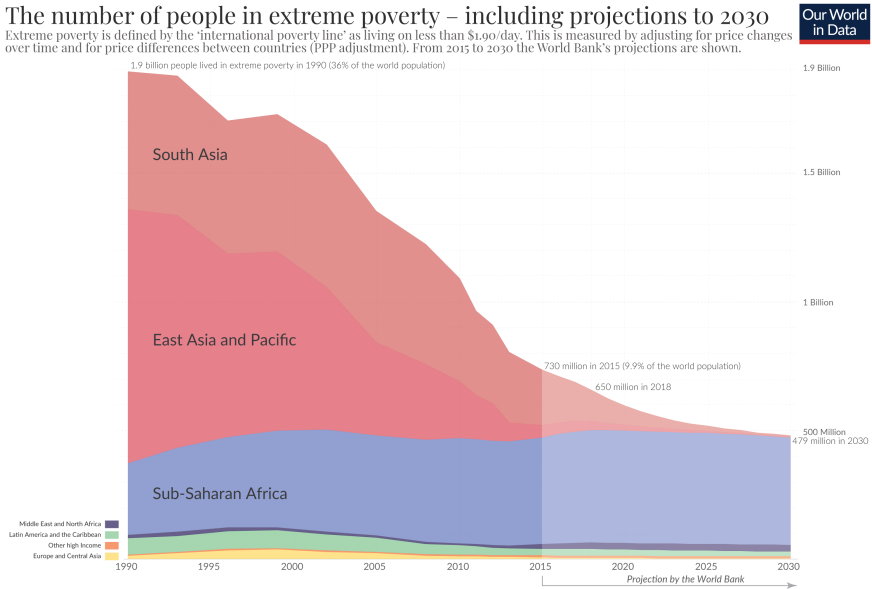


Data source: World Bank Poverty and Inequality Platform (2024)

OurWorldInData.org/poverty | CC BY

Note: This data is expressed in international-$[1] at 2017 prices. Depending on the country and year, it relates to income measured after taxes and benefits, or to consumption, per capita[2].

Poverty eradication



Data source: World Bank data from 1990 to 2015. The projections from 2015 to 2030 are published in the World Bank report *Poverty and Shared Prosperity 2018*.
This is a visualization from OurWorldinData.org, where you find data and research on how the world is changing.

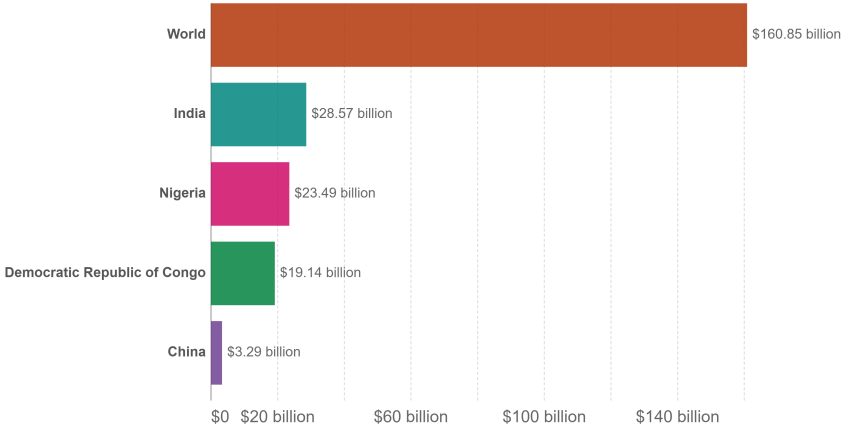
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Poverty eradication

The poverty gap, in international-$, 2013

The poverty gap is the amount of money that would be theoretically needed to lift the incomes of all people in extreme poverty up to the international poverty line of $1.90 a day. These estimates are expressed in international dollars using 2011 PPP conversion rates. This means that figures account for differences in prices levels, as well as for inflation.

Our World in Data

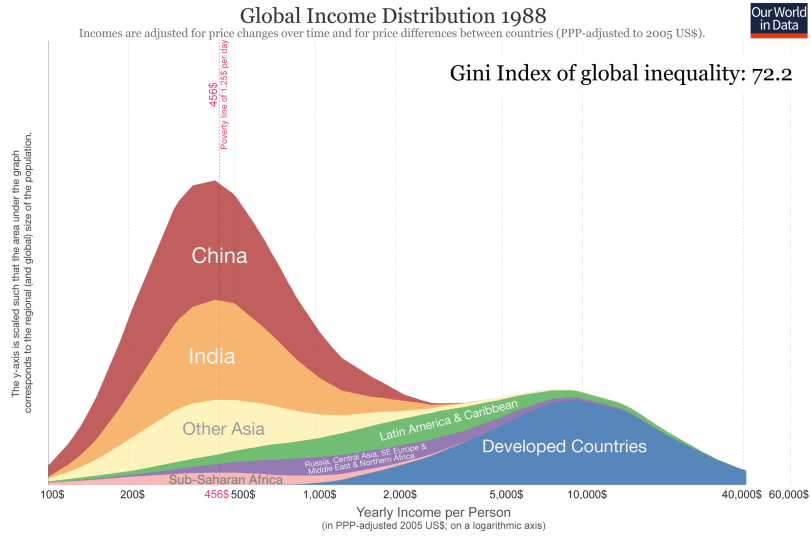


Source: PovcalNet (World Bank) (2017)

OurWorldInData.org/extreme-poverty/ • CC BY

Note: The cost of closing the poverty gap does not take into account costs and inefficiencies from making the necessary transfers.

Poverty eradication



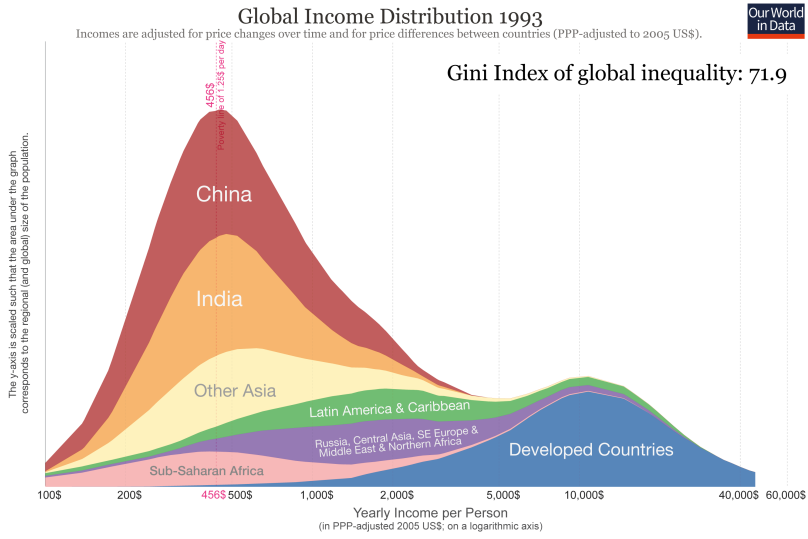
Data source: Lakner and Milanovic (2015) – *Global Income Distribution: From the Fall of the Berlin Wall to the Great Recession*, World Bank Economic Review.

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The categorisation of countries is stable over the entire time period 1988-2011.

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Poverty eradication



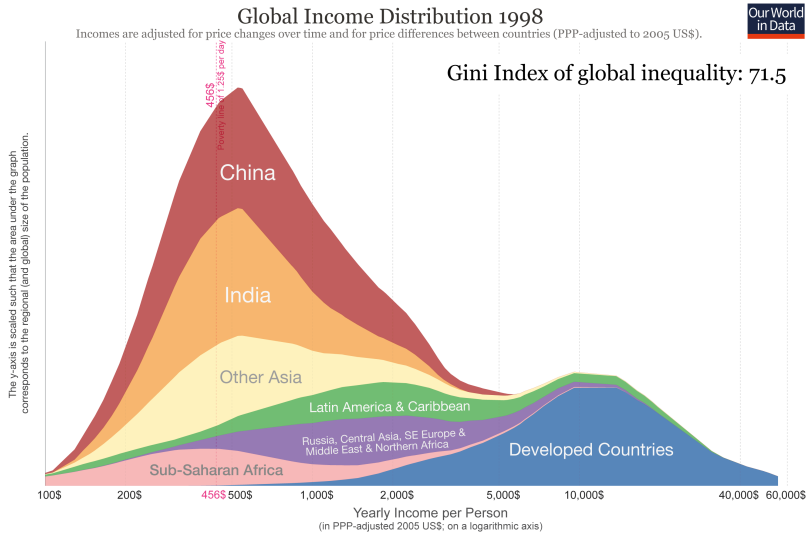
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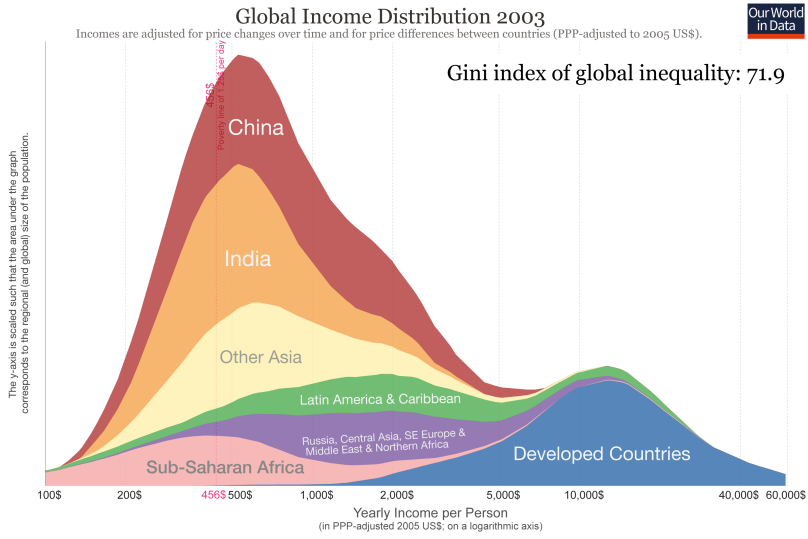
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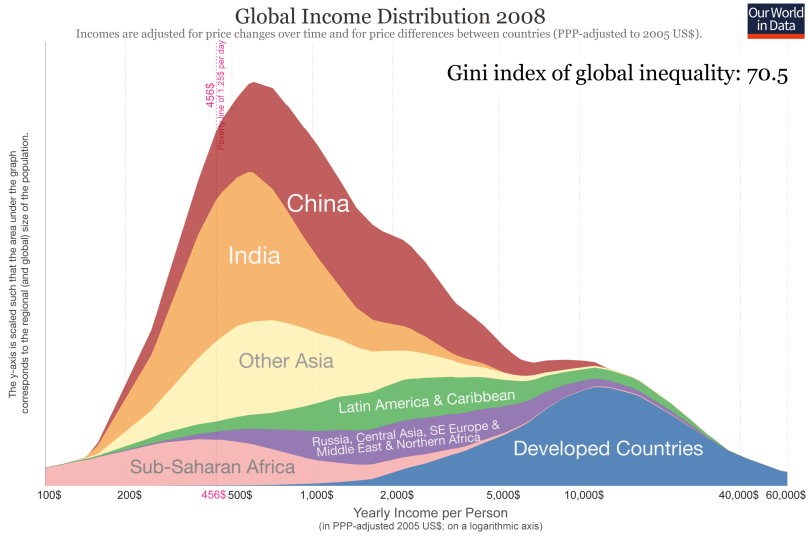
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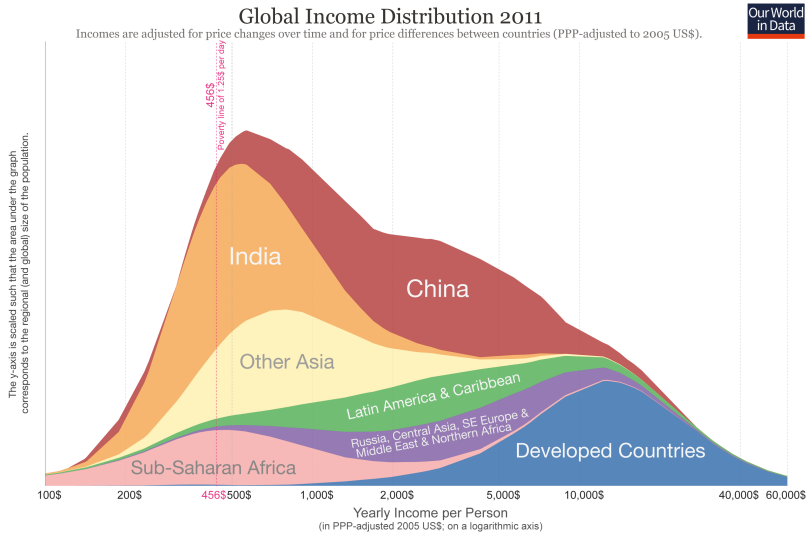
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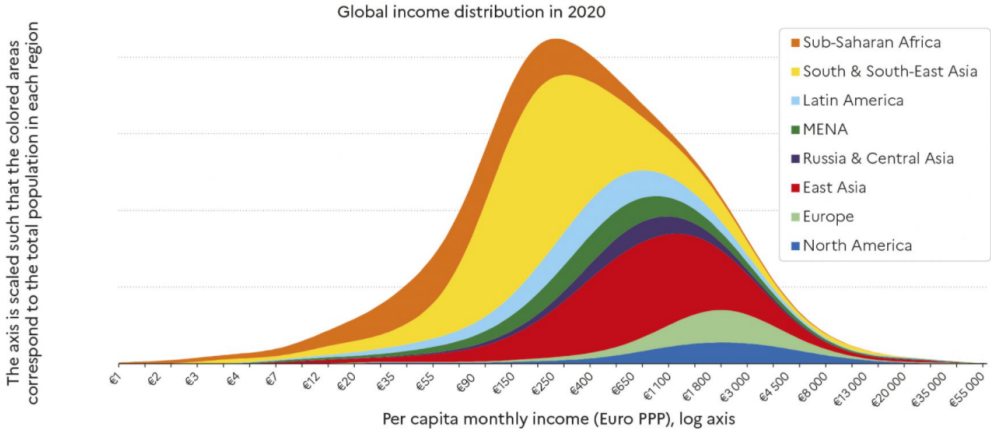
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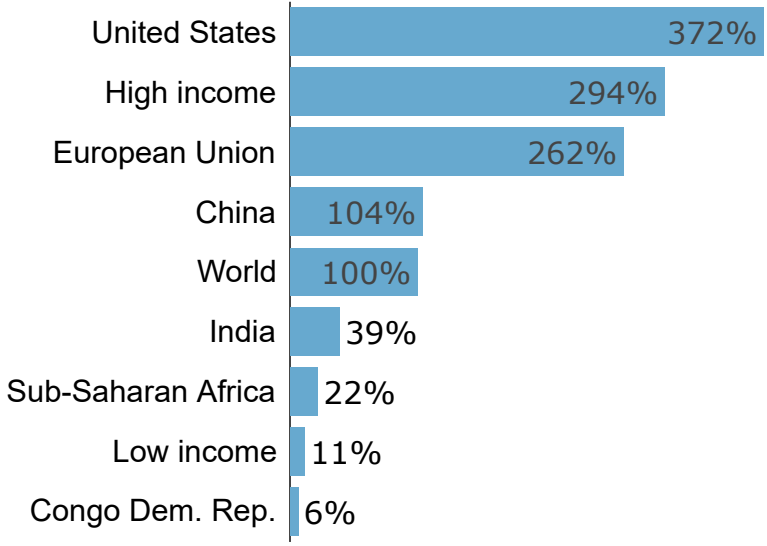
Poverty eradication



Interpretation: *The graph shows the size and geographical repartition of the global population at different levels of the income distribution. The relative size of each color wedge is proportional to the relative population in a region. Incomes are measured after pension and unemployment insurance transfers and before income and wealth taxes.* ***Note:*** *Distribution of per capita incomes (for the distribution of per-adult incomes, see Chapter 1).* ***Sources and series:*** *wir2022.wid.world/methodology.*

Poverty eradication

GDP per capita in PPP, relative to world average (2023).



Poverty eradication

Extreme poverty := $2.15 a day (in 2017 PPP $) ≈ threshold of undernourishment.

Severe poverty := $3.65/day. *Acute* poverty := $6.85/day.

SDG 1 (2015): Eradicate extreme poverty by 2030.

Can we eradicate extreme poverty by 2030?

⇒ Let's use idealized redistributive policies to assess each country's capacity to end poverty.

Methodology

Idealized redistributive policies

Tax policies that would raise enough revenues to eradicate poverty.

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⚠These policies are idealized (assume administrative capacity at no cost, no distorsion) ⇒ upper bound of what they could achieve.

Literature

Ravallion (2010): a 100% tax above the U.S. poverty line insufficient to eradicate extreme poverty in 29 countries.

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More than 60 countries expected to fail SDG 1 (Moyer & Hedden, 2020).

Primer of the results: comparison with BCL

$\Rightarrow$ 3 contributions w.r.t. BCL:

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In 2030 after 3% growth: figures go down to 34 and 6 countries.

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⇒ As robustness check, I impute extra consumption in HFCE to top percentile (as Lakner & Milanovic, 2015; Anand & Segal, 2015).

Growth scenarios

Consistent with the literature, I assume balanced growth: each percentile grows at same rate between the survey year and 2030.

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7% growth since 2016 as per SDG 8.1.

The effects of balanced growth

Table 1: Global poverty rates and poverty gaps in 2030 under different growth scenarios. Poverty rates are expressed in % of world population and poverty gaps in % of world GDP. Poverty lines are in 2017 PPP $/day.

Growth scenario	Poverty rate (%)				Poverty gap (% of GDP)			
(Poverty line in $/day)	2.15	3.65	6.85	18.15	2.15	3.65	6.85	18.15
2022 Estimate	7.3	21.1	44.4	72.2	0.26	1.36	7.01	42.96
Trend (2014–2019)	6.2	14.4	34.5	66.2	0.21	0.87	4.29	30.64
Max(Trend, 0)	6.3	14.2	34.3	66.4	0.19	0.81	4.16	30.25
Autoregressive projection	6.2	15.2	36.8	65.5	0.17	0.84	4.64	32.02
3% growth	5.2	15.2	37.5	68.2	0.14	0.75	4.38	31.20
3% unbalanced growth	5.1	15.1	37.5	67.7	0.19	0.83	4.48	31.83
7% growth	2.2	8.5	25.5	59.5	0.05	0.29	1.93	18.07
7% growth since 2016	1.1	3.1	15.3	51.3	0.01	0.08	0.74	10.15

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Trend (2014–2019)	6.2	14.4	34.5	66.2	0.21	0.87	4.29	30.64
Max(Trend, 0)	6.3	14.2	34.3	66.4	0.19	0.81	4.16	30.25
Autoregressive projection	6.2	15.2	36.8	65.5	0.17	0.84	4.64	32.02
3% growth	5.2	15.2	37.5	68.2	0.14	0.75	4.38	31.20
3% unbalanced growth	5.1	15.1	37.5	67.7	0.19	0.83	4.48	31.83
7% growth	2.2	8.5	25.5	59.5	0.05	0.29	1.93	18.07
7% growth since 2016	1.1	3.1	15.3	51.3	0.01	0.08	0.74	10.15

Among the 8 countries < $3/day; 2014–19 growth on average negative, highest growing rate at 2.4%.
⇒ My benchmark of 3% growth over 2022–30 is optimistic.

Antipoverty caps

Income cap eradicating extreme poverty (in $/day), after growth of 3% over 2022–2030.



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In a very optimistic scenario of 7% growth, the anti-extreme-poverty cap would be $14/day in the D.R.C and $8.60 in Burundi.

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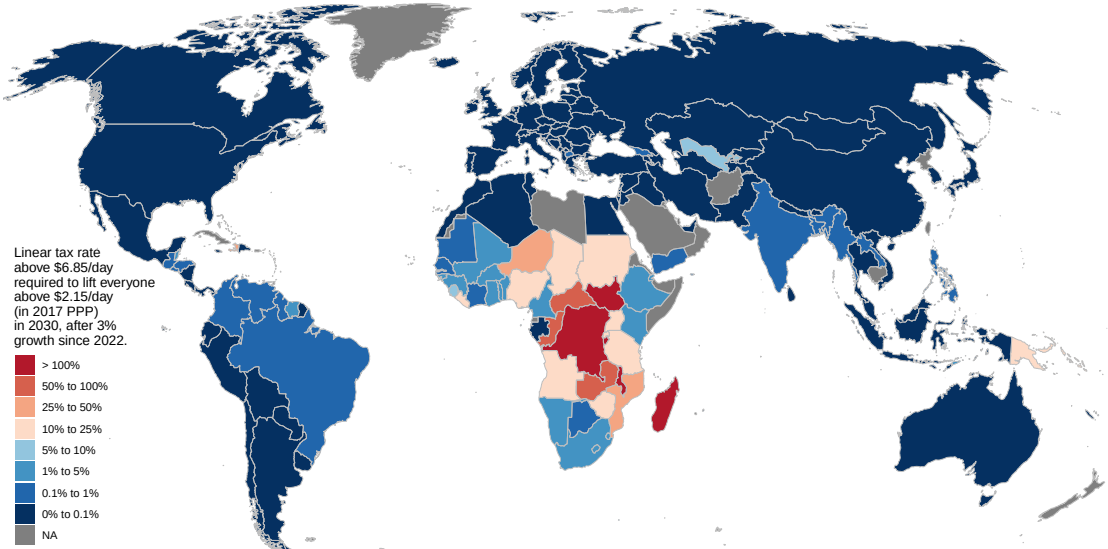
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With BCS ($4.35 in median), poverty more prevalent:
Despite lot of missing data (e.g. India, D.R.C.), 14 countries' average consumption < BCS in 2030 after 3% growth (incl. Nigeria).

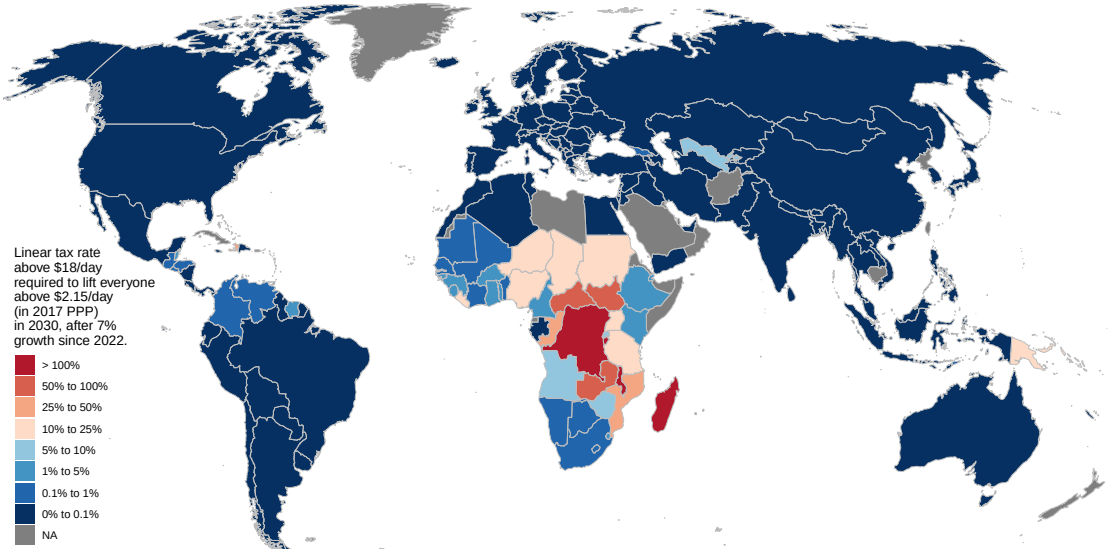
Antipoverty taxes

Linear tax rate above $6.85/day eradicating extreme poverty (in %), after a growth of 3% over 2022–2030.



Antipoverty taxes

Linear tax rate above $18.15/day eradicating extreme poverty (in %), after a growth of 7% over 2022–2030.



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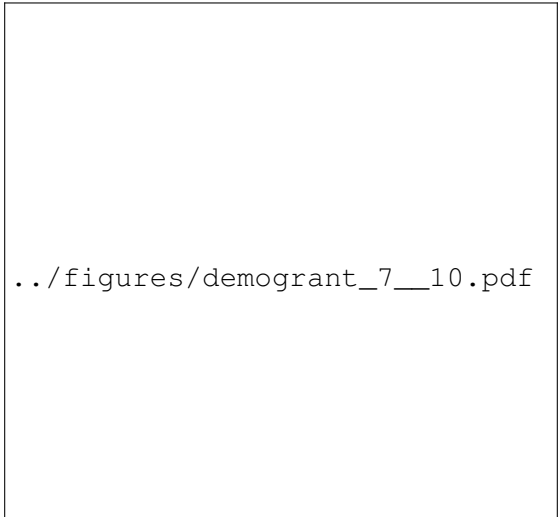
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In India, the anti-extreme-poverty tax above $18.15/day is:

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- 156% after 3% growth

Fundable income floor

Income floor that can be funded with a 10% marginal tax on income above $6.85/day, after growth of 3% over 2022–2030.



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Extreme poverty gap (0.17% of global real GDP) $\lesssim$ ODA shortfall (0.2% of global nominal GDP).

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A wealth tax at 4% above $1 million and 10% above $100 million would raise even more while targeting the top 1%.

Conclusion

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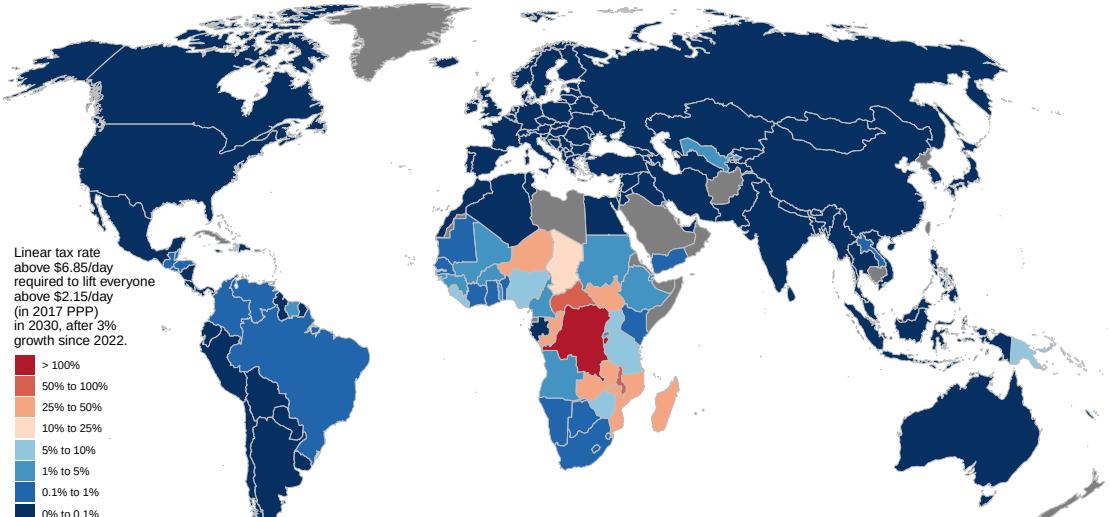
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Thank you for your attention!

Read paper on SSRN: 4725417, and for more: @adrien_fabre.

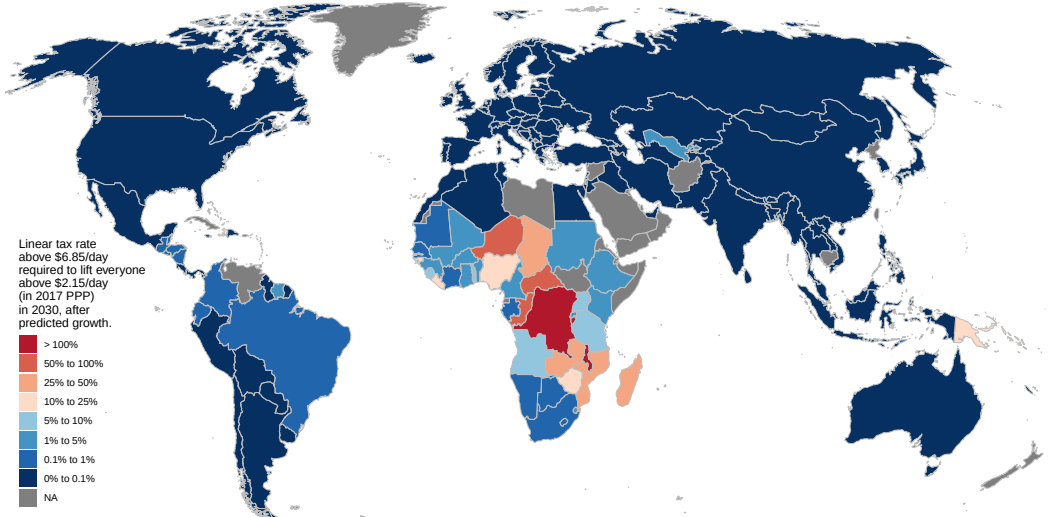
Antipoverty taxes

Linear tax rate above $18.15/day eradicating extreme poverty (in %), after a growth of 7% over 2022–2030, HFCE-scaled.



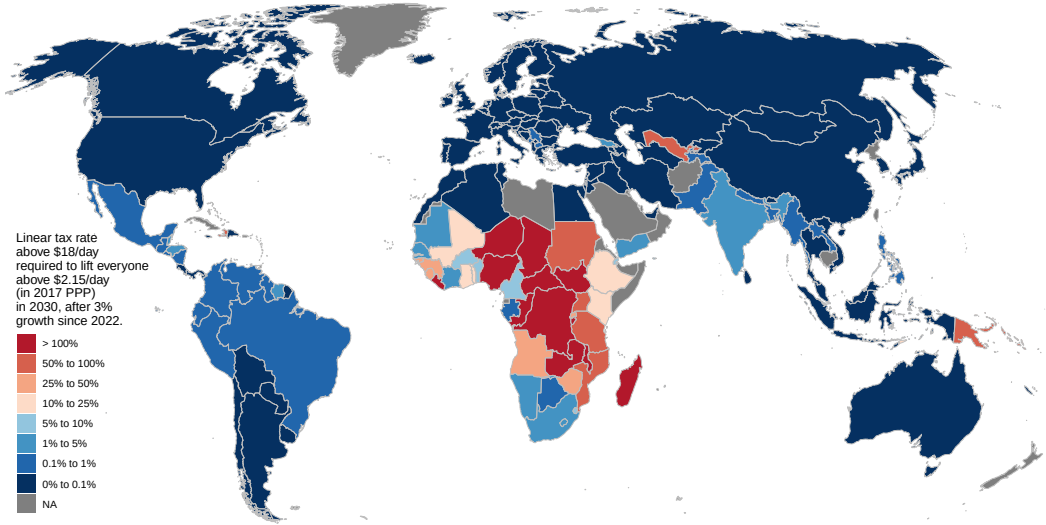
Antipoverty taxes

Linear tax rate above $18.15/day eradicating extreme poverty (in %), with growth until 2030 predicted at the country level.



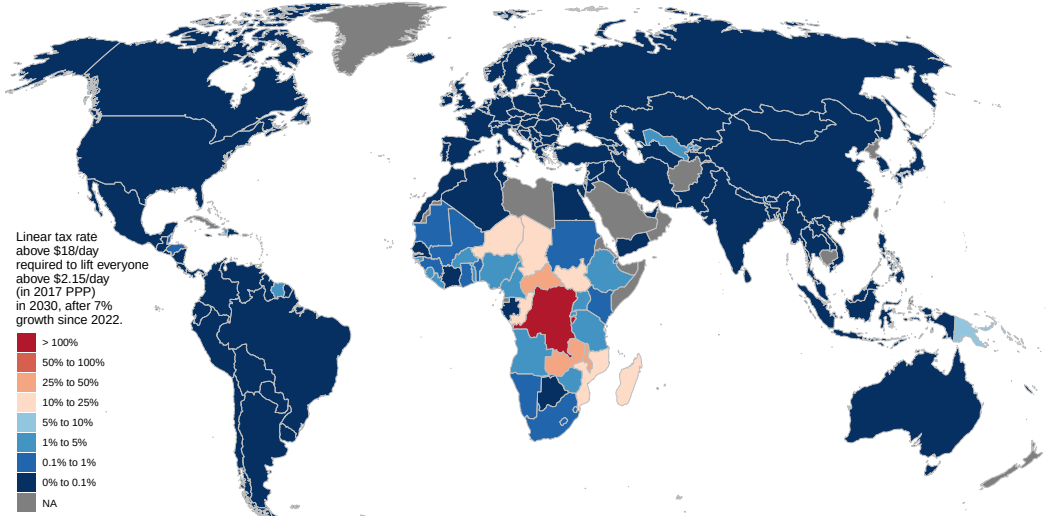
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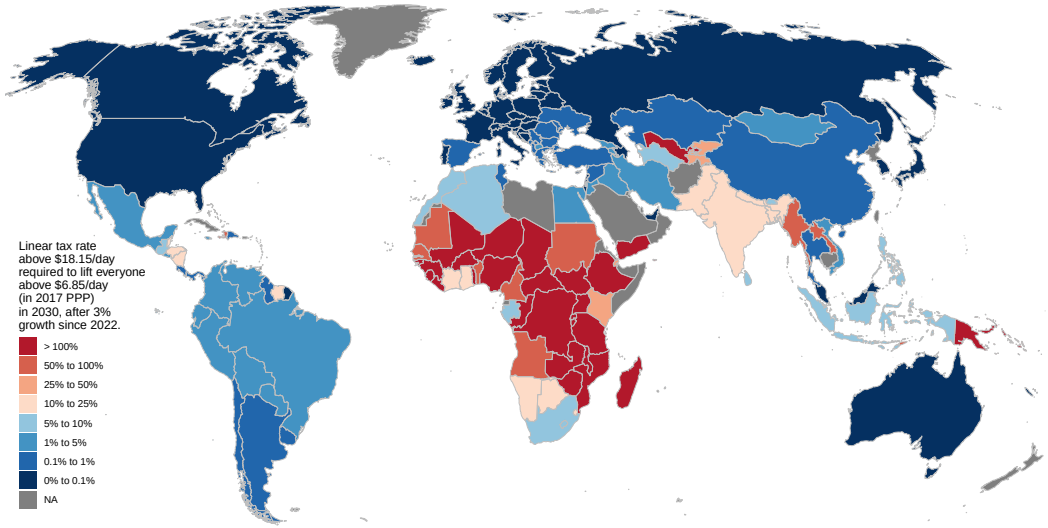
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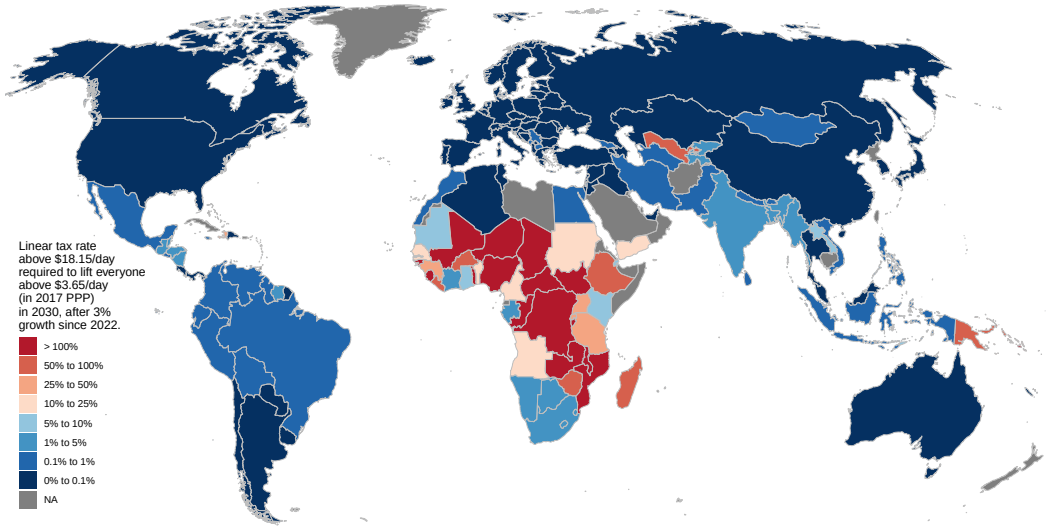
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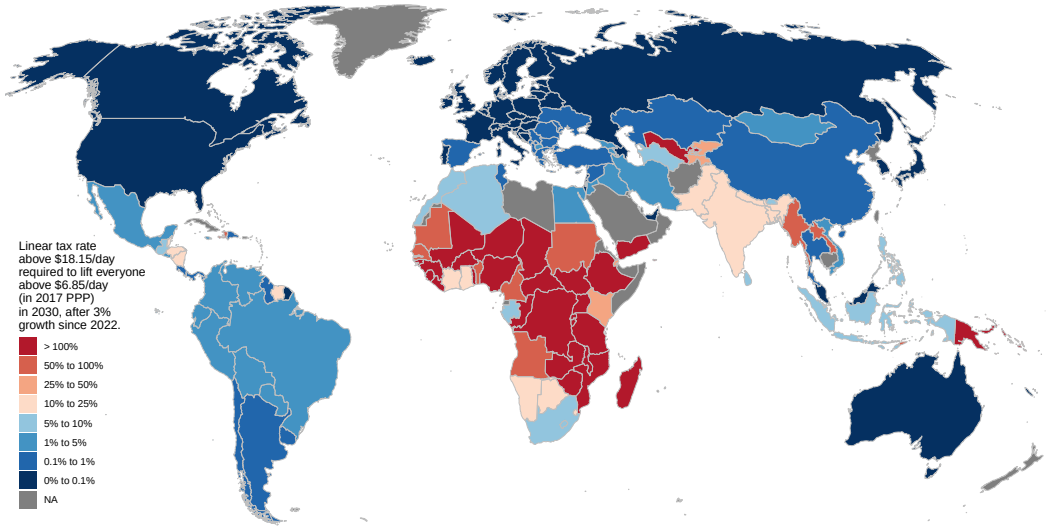
Antipoverty taxes

Linear tax rate above $18.15/day eradicating severe poverty (in %), after a growth of 3% over 2022–2030.



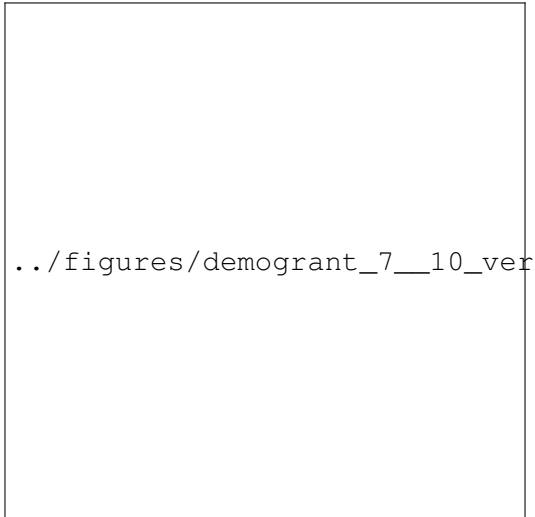
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